

MOHAMED AYMEN AKCHICHE

FULL STACK AI ENGINEER

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PROFESSIONAL SUMMARY

Early-career Full Stack AI Engineer combining software engineering fundamentals with a focus on building full-stack applications with generative AI and agentic capabilities embedded into the solution architecture. Computer Science Engineering graduate specializing in AI, with experience across full-stack web architectures (React, Node.js, FastAPI, REST APIs, SQL) and applied AI systems (LLM APIs, LangChain, LangGraph, RAG, vector databases). Demonstrates judgment in determining when AI creates measurable value versus when traditional software is appropriate. Adept at using AI-assisted coding tools while maintaining ownership and deep understanding of production code.

EDUCATION

State Engineer Degree in Artificial Intelligence

Kasdi Merbah University of Ouargla

5-Year Engineering Program | Computer Science & AI Specialization

Bachelor Foundation: Software engineering, algorithms, databases, data structures, and application development.

TECHNICAL SKILLS & CAPABILITIES

- **Core Engineering & Languages:** Python, JavaScript, TypeScript, REST APIs, Git, SQL, Debugging, Testing, Application Architecture.
- **Full Stack Web Development:** React, Node.js, Express.js, FastAPI, Frontend/Backend Integration, Database Management.
- **Generative & Agentic AI:** LLM APIs, LangChain, LangGraph, RAG, Vector Databases (ChromaDB), Embeddings, Tool Calling, Conversational Memory, Agentic Workflows.
- **Applied AI / ML & Analytics:** Scikit-learn, NLTK, Sentence Embeddings, Data Preprocessing, Predictive Analytics, Model Evaluation.
- **Deployment & Cloud Concepts:** Streamlit, FastAPI Deployment, Web Application Deployment, Cloud Platforms (AWS/GCP exposure).
- **AI-Assisted Development:** Claude, ChatGPT, Codex; AI-assisted code acceleration with thorough validation, debugging, and code ownership.

FEATURED FULL-STACK & AI PROJECTS

AI-Powered E-Learning Multi-Agent Platform

Python, LangChain, LangGraph, Gemini, ChromaDB, RAG, React/JS

- * Architected an end-to-end multi-agent AI assistant integrating LLM logic with structured databases, specialized tools, and role-aware enterprise workflows.
- * Engineered a Main Agent for request routing, dynamic tool calling, and context synthesis across student, course, and exam datasets.
- * Implemented RAG with embeddings and ChromaDB alongside a History Agent for conversational memory fallback and failure scenario handling.

Job Scoring & Screening Multi-Agent System

Python, LLM APIs, RAG, ChromaDB, LangChain/LangGraph

- * Designed an agentic system converting job requirements into structured evaluation criteria and screening questions.
- * Built document-grounded evaluation using targeted RAG retrieval and structured JSON outputs to deliver measurable candidate scoring.

Full-Stack Web & AI Development

Freelance / Independent | JS, React, Node.js, FastAPI, REST APIs, SQL

- * Built full-stack applications connecting modern React frontends to robust Node.js/FastAPI backends and relational databases via clean REST APIs.
- * Applied software engineering best practices including version control (Git), API design, modular architecture, and thorough debugging.

AI Development Internship

Somu Raqami | Python, FastAPI, LangChain, RAG, LLM APIs

- * Contributed to enterprise AI solutions incorporating FastAPI, LangChain workflows, text embeddings, and vector store retrieval.
- * Iterated on application failure modes and response performance, improving message-classification accuracy.

Predictive Systems & AI Decision Tools

Python, Scikit-learn, Pandas, Gemini, Streamlit

- * Built end-to-end client applications combining predictive Machine Learning models with LLM-generated actionable business recommendations.

LEADERSHIP & LANGUAGES

- **Course Instructor:** Scientific Corner Scientific Club. Designed programming curriculum, mentored students, and communicated technical concepts clearly to diverse learners.
- **Languages:** Arabic (Native), English (Professional Technical Proficiency).